2312220 6 Sec A

M. Abdullat

Outlook	T	Hum	W	P
- Sunny	- Hot	- High	- Weak	No
- Sunny	+ Mild	- High	+ Strong	No
+ Cloudy	- Hot	- High	- Weak	Yes
+ Rainy	+ Mild	- High	- Weak	Yes
+ Rainy	+ Cool	+ Normal	- Weak	Yes
+ Rainy	+ Cool	+ Normal	+ Strong	No
+ Cloudy	+ Mild	+ Normal	+ Strong	Yes
- Sunny	+ Cool	+ Normal	- Weak	No

(1) Entropy Calculation of Prime Attribute

$$H(S) = -\frac{3}{8}\log_2\frac{3}{8} - \frac{2}{8}\log_2\frac{2}{8}$$

$$H(S) = 1.56$$

(2) Gain Info:

gain info $(S, A) = H(S) - H(S/A)$

OF Outlook -

Outlook	Total	Yes	No	Entropy
Sunny	3	0	3	0
Cloudy	2	2	0	0.0
Rainy	3	2	1	0.91

$$-\frac{0}{3}\log_2\frac{0}{3} - \frac{3}{3}\log_2\frac{3}{3}$$

* The weighted Entropy

$H(S/OL) = \frac{3}{8}(0.00) + \frac{2}{8}(0.00) + \frac{3}{8}(\text{illegible})$

$H(S/OL) = 0.341$

* Gain Info $(S, OL) = 1.56 - 0.341$

$= 1.219$

OF ~~Hot~~ temperature-

T	Total	Yes	No	Entropy
Hot	2	1	1	1
Mild	3	2	1	0.91
Cool	3	1	2	0.91

* The wighted Entropy

$H(S/T) = \frac{2}{8}(1) + \frac{3}{8}(0.9) + (0.91)\frac{3}{8}$

$H(S/T) = 0.9385$

* Gain Info $(S, T) = 1.56 - 0.9385$

$= 0.6215$

of Humidity

H	Total	Yes	No	Entropy
High	4	2	2	1
Normal	4	2	2	1

* The weighted Entropy

$$H(S/H) = \frac{4}{8}(1) + \frac{4}{8}(1)$$

$$H(S/H) = 1$$

* Gain Infor $(S, H) = 1.56 - 1$

$$= 0.56$$

of Wind

W	Total	Yes	No	Entropy
Weak	5	3	2	0.97
Strong	3	1	2	0.918

* The Weighted Entropy

$$H(S/W) = \frac{5}{8}(0.97) + \frac{3}{8}(0.918)$$

$$= 0.95$$

* Gain Info $(S, W) = 1.56 - 0.95$

$$= 0.61$$

* Compare Info Gain

Attribute	Weighted Ent	Info g.
OL	0.391	1.219
T	0.938	0.625
H	1	0.56
W	0.95	0.61

→ We will choose OL because it has highest info gain

